

# ESS INDIA Tools and Technologies

## ESS India Tools and Technologies

Here is the comprehensive technology and tools stack used to build the **ESS India Portal & CMS**:

---

### Core Platform

| Category        | Technologies            |
|-----------------|-------------------------|
| Language        | TypeScript, JavaScript  |
| Runtime         | Node.js 20.x / 22.x     |
| Framework       | Next.js 16 (App Router) |
| Frontend        | React 19                |
| Package Manager | npm 10.x                |
| Version Control | Git (GitHub)            |

---

### Frontend / UI

| Category              | Technologies                                  |
|-----------------------|-----------------------------------------------|
| Styling               | Tailwind CSS 4, PostCSS, Autoprefixer         |
| UI Components         | Radix UI (Base UI), shadcn/ui components      |
| Icons                 | Lucide React, flag-icons (country selections) |
| Animations            | Framer Motion, tw-animate-css                 |
| Forms                 | React Hook Form, Zod, Hookform Resolvers      |
| State / Data Fetching | TanStack React Query (v5)                     |

|                            |                                                               |
|----------------------------|---------------------------------------------------------------|
| <b>Rich Text Editor</b>    | Tiptap Editor (Starter-Kit, Link, Underline, Placeholder, PM) |
| <b>Other UI Components</b> | Sonner (toasts), next-themes (theme management)               |

---

## Backend / API

| Category                    | Technologies                                                                         |
|-----------------------------|--------------------------------------------------------------------------------------|
| <b>API Layer</b>            | Next.js API Route Handlers ( <code>src/app/api/*</code> )                            |
| <b>Validation</b>           | Zod schemas                                                                          |
| <b>Auth</b>                 | Custom JWT Token Auth, Session Middleware                                            |
| <b>Database Client</b>      | Drizzle ORM ( <code>drizzle-orm</code> ) & Postgres Client ( <code>postgres</code> ) |
| <b>BaaS / DB Platform</b>   | Supabase Hub ( <code>@supabase/supabase-js</code> , <code>@supabase/ssr</code> )     |
| <b>Cache / Redis Client</b> | ioredis                                                                              |
| <b>Email Gateway</b>        | Nodemailer (SMTP Relay / Mailgun / etc.)                                             |

---

## Database & Schema

| Category                  | Technologies                                                              |
|---------------------------|---------------------------------------------------------------------------|
| <b>Database</b>           | PostgreSQL (hosted on Supabase)                                           |
| <b>Schema Management</b>  | Drizzle Kit migrations ( <code>drizzle-kit generate/migrate/push</code> ) |
| <b>Connection Pooling</b> | Supabase connection pooler / transaction pooler                           |

---

## File Handling & Exports

| Category              | Technologies                                                                               |
|-----------------------|--------------------------------------------------------------------------------------------|
| <b>Asset Storage</b>  | AWS S3 Bucket / Supabase Storage ( <code>@aws-sdk/client-s3</code> , S3 Request Presigner) |
| <b>DOCX Parsing</b>   | Mammoth (converts applicant DOCX files to HTML)                                            |
| <b>PDF Extraction</b> | Unpdf (extracts text content from uploaded resumes)                                        |
| <b>PDF Generation</b> | jsPDF (generates client-ready application/leads summaries)                                 |

---

## Infrastructure & Hosting

| Category               | Technologies                                                    |
|------------------------|-----------------------------------------------------------------|
| <b>Hosting</b>         | VPS Application Server (Ubuntu)                                 |
| <b>Reverse Proxy</b>   | Nginx reverse proxy                                             |
| <b>Process Manager</b> | PM2 Process Daemon Manager ( <code>ecosystem.config.js</code> ) |
| <b>Asset CDN</b>       | AWS CloudFront / Cloudflare Edge                                |
| <b>TLS / DNS</b>       | Let's Encrypt SSL Certbot + Domain DNS                          |

---

## DevOps & Operations

| Category              | Technologies                                            |
|-----------------------|---------------------------------------------------------|
| <b>Build Pipeline</b> | Next.js Production Build ( <code>npm run build</code> ) |

|                              |                                                            |
|------------------------------|------------------------------------------------------------|
| <b>Backup Scripts</b>        | Automated pg_dump database scripts + cron backups          |
| <b>Monitoring &amp; Logs</b> | PM2 service logs ( pm2 logs ) & Nginx access/error logs    |
| <b>Containerization</b>      | Docker, Docker Compose ( Dockerfile , docker-compose.yml ) |

---

#### Analytics & Utilities

| Category                 | Technologies                                   |
|--------------------------|------------------------------------------------|
| <b>Date Handling</b>     | Native Date utilities                          |
| <b>Class Utilities</b>   | clsx, tailwind-merge, class-variance-authority |
| <b>Dial Code Mapping</b> | country-codes-list                             |

---

#### Application Modules (Functional Areas)

- **Dynamic Page Creation Wizard:** Multi-step manager to name paths, configure SEO metadata, and preview drafts before publishing.
- **Component-Based Layout Engine:** Interactive block builder utilizing pre-configured template sections (Hero components, FMCG use-cases, testimonials list, HospitalHero grids).
- **Multi-Behavior Header Navigation:** Edit links and drop-down layouts:
  - *Simple Link:* Navigates directly to target URL.
  - *Vertical Dropdown:* Compact vertical list of sub-links managed dynamically inside the form.
  - *Mega Menu Grid:* Double-column horizontal layouts populated via Category Trees.
- **Global & Page-Specific SEO Controllers:** Configures Title, Meta Description, canonical paths, og:image properties, and JSON-LD schema markup dynamically.
- **Media Assets Library:** Handles visual uploads directly to S3 cloud storage buckets.
- **Applicant Processing System:** Receives applications, validates inputs (email and country-specific phone dial formats), parses resume files (PDF/DOCX), and triggers outbound notifications.

- **Client Inquiry Lead Captures:** Captures website interactions, processes country dial verification rules, and outputs entries to the admin panel.